QUBO-Based Hybrid LDPC Decoding for Coherent Optical Optimization

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***Abstract*—Low-density parity-check (LDPC) decoding can be expressed as a discrete optimization problem that combines channel likelihoods with parity constraints. This paper organizes a competition project on hybrid classical and coherent-optical-oriented decoding into a unified modeling framework. The decoder uses classical initialization, reliability-based subproblem selection, quadratic unconstrained binary optimization (QUBO), and likelihood-based candidate selection. An explicit integer-slack formulation makes the parity penalty quadratic, while boundary syndromes preserve constraints when only part of a codeword is optimized. The formulation also provides a transparent count of data and auxiliary variables under a limited solver budget. The two regular-code scenarios in the competition are distinguished from standardized 5G NR base graphs. Numerical values and plots from the archived report are retained as reported project results, with their provenance and reproducibility limits stated explicitly. Local checks verify the clarified parity model and code dimensions; they do not constitute a new communication-performance benchmark. The study provides a structured account of the project and its mathematical model without inferring physical quantum advantage from SDK simulation.**

***Index Terms*—LDPC codes, QUBO, hybrid decoding, coherent optical optimization, simulated annealing, maximum likelihood.**

I. Introduction

Low-density parity-check codes represent linear error-correcting codes by sparse parity-check matrices. Their sparse graphical structure supports efficient iterative decoding and has made them central to modern communication systems [1]–[3]. Nevertheless, finite-length codes can exhibit difficult error patterns, and the tradeoff between decoding reliability, latency, and computational resources remains important. Optimization-based decoding offers a complementary way to search for parity-consistent codewords using the information supplied by a noisy channel.

QUBO models connect this decoding problem to annealing and Ising optimization. Kasi and Jamieson studied quantum belief propagation for LDPC decoding in wireless networks [4]; Guo *et al.* investigated quantum annealing with maximum-likelihood post-processing [5]. These works motivate combining classical inference with optimization rather than treating the optimizer as a complete communication receiver. A QUBO representation is a modeling interface: it does not, by itself, specify whether the computation is classical simulation, coherent optical processing, or quantum annealing.

The project addressed Problem B of the Third Quantum Information Technology and Application Innovation Competition, in the mathematical modeling track based on coherent optical quantum computers. The task required an LDPC-to-QUBO conversion, use of a simulated solver in the Kaiwu SDK, evaluation at multiple signal-to-noise ratios (SNRs), and an approach to longer codes under limited computational resources. Its two illustrative scenarios use regular $(2, 3)$ and $(3, 6)$ codes. The problem's references to BG1 and BG2 provide an application motivation; they do not make these regular ensembles equivalent to the standardized 5G NR constructions.

The project combines four ideas: a channel-aware QUBO objective; selection of unreliable bits and neighboring constraints; repeated optimization followed by likelihood-based selection; and decomposition for a limited solver budget. This English edition consolidates repeated material in the final Chinese report, standardizes notation, and makes the intended parity model explicit. Section III states the clarified mathematical formulation, Section IV describes the proposed architecture, and Section V preserves the numerical record with a distinction between reported values and independently checked properties.

Scope of this edition: This is an English reorganization of an archived competition report, prepared for author review. Historical numerical values are not presented as newly reproduced experiments. Mathematical clarifications and verification performed for this edition are identified separately; they do not imply that the archived implementation already contains every clarified step. The original project files are preserved.

II. Problem Definition and Code Conventions

A. Code Length, Dimension, and Rate

Let $H \in \mathbb{F}_2^{m\times n}$ be a parity-check matrix. A codeword $x \in \mathbb{F}_2^n$ belongs to

$$\begin{aligned} \mathcal{C} &= \{x \in \mathbb{F}_2^n : Hx = 0\}, \\ K &= n - \operatorname{rank}_{\mathbb{F}_2}(H), \qquad R = K/n. \end{aligned} \tag{1}$$

The number of matrix rows is not necessarily the number of independent checks. Consequently, $1 - m/n$ is a design rate; it equals the actual rate only when H has full row rank. Information length K, transmitted codeword length n, and the number of check equations m must be reported separately.

Throughout this paper, a regular (j, k) code has column degree j and row degree k. The number of edges satisfies $nj = mk$, giving a design rate $1 - j/k$. Thus the two regular families have design rates $1/3$ and $1/2$, respectively. The original problem reverses the verbal degree description

TABLE I
COMPETITION SCENARIOS AND DIMENSIONAL INTERPRETATION

Item	Throughput-oriented	Complexity-oriented
Regular family	$(2, 3)$	$(3, 6)$
Source wording	180 input bits	120 input bits
Design rate	$1/3$	$1/2$
If number means n	$H : 120 \times 180$	$H : 60 \times 120$
If number means K	$n \approx 540$	$n \approx 240$
Application label	BG1-oriented	BG2-oriented

The last interpretation uses the design rate; the actual dimension must be determined from the generated matrix. The scenario labels are inherited from the problem, not 3GPP conformance claims.

while using these familiar labels; the separate English problem statement preserves and explains that ambiguity.

B. Relationship to 5G NR

5G NR specifies two quasi-cyclic LDPC base graphs, BG1 and BG2 [6]. Their base-matrix dimensions are 46×68 and 42×52. With lifting factor Z, the corresponding full lifted matrices have $68Z$ and $52Z$ columns before the prescribed puncturing and rate-matching operations. The maximum code-block sizes associated with the information-side construction are 8448 and 3840 bits. These quantities should not be interchanged with the full mother-codeword length.

The archived code includes regular sparse matrices and simplified randomly constructed lifted matrices. A random matrix having a base graph's shape is not the standardized base graph: standard shift coefficients, lifting rules, filler bits, puncturing, segmentation, and rate matching are also required. The project therefore uses BG1/BG2 terminology as an application-oriented label unless a particular standards-compliant construction is explicitly established. Hardware capacity must likewise be expressed as an actual configurable solver limit, rather than transferring a capacity quoted for another platform.

C. Channel and Decoding Objective

For binary phase-shift keying (BPSK), the transmitted symbol is $s_i = 1 - 2x_i$, and the real additive white Gaussian noise (AWGN) channel is

$$y_i = s_i + w_i, \qquad w_i \sim \mathcal{N}(0, \sigma^2). \tag{2}$$

The channel log-likelihood ratio (LLR) is defined consistently as

$$L_i = \log \frac{p(y_i \mid x_i = 0)}{p(y_i \mid x_i = 1)} = \frac{2y_i}{\sigma^2}. \tag{3}$$

Positive L_i favors zero. For equiprobable codewords and a memoryless channel, maximum-likelihood (ML) decoding is therefore

$$\hat{x}_{\mathrm{ML}} = \arg\min_{x \in \mathcal{C}} \sum_{i=1}^{n} L_i x_i. \tag{4}$$

The positive sign in this minimization follows from the LLR definition; maximizing the same sum would reverse the preference.

The SNR convention must be stated when generating noise. With unit symbol energy and $\gamma_s = E_s/N_0$, $\sigma^2 = 1/(2\gamma_s)$. If the horizontal axis instead uses $\gamma_b = E_b/N_0$ per information bit, then $\gamma_s = R\gamma_b$ and $\sigma^2 = 1/(2R\gamma_b)$. A dB value alone is insufficient to distinguish these conventions.

III. QUBO Modeling of LDPC Decoding

A. Exact Quadratic Parity Penalty

The intended objective combines channel cost with penalties for invalid parity checks. Directly writing a modulo operation in an energy function does not yet yield a quadratic polynomial. An explicit construction is needed before submitting the model to a QUBO solver [7].

For check a, let $\mathcal{N}(a)$ be the participating variable indices, $d_a = |\mathcal{N}(a)|$, and $b_a \in \{0, 1\}$ the target parity. Introduce a nonnegative integer t_a such that

$$\sum_{i \in \mathcal{N}(a)} x_i - b_a - 2t_a = 0. \tag{5}$$

For a nonempty check, an exact binary representation is

$$t_a = \sum_{r=0}^{q_a - 1} 2^r z_{a,r}, \qquad z_{a,r} \in \{0, 1\}, \tag{6}$$

$$q_a = \left\lceil \log_2 \left(\left\lfloor \frac{d_a - b_a}{2} \right\rfloor + 1 \right) \right\rceil. \tag{7}$$

Zero bits are used when $q_a = 0$. The binary representation may allow slack values larger than the largest needed value. They cannot create spurious zero-penalty assignments because equality in (5) still determines t_a exactly. An empty check with $b_a = 0$ can be discarded; an empty check with $b_a = 1$ is infeasible and must be detected.

The resulting unconstrained binary objective is

$$\begin{aligned} E(x, z) &= \sum_{i=1}^{n} L_i x_i + \sum_{a=1}^{m} A_a \rho_a(x, z)^2, \\ \rho_a(x, z) &= \sum_{i \in \mathcal{N}(a)} x_i - b_a - 2 \sum_{r=0}^{q_a - 1} 2^r z_{a,r}. \end{aligned} \tag{8}$$

Each expression inside the square is linear in binary variables. Since $u^2 = u$ for $u \in \{0, 1\}$, expansion produces only constant, linear, and pairwise terms. For a complete codeword, $b_a = 0$ for every check. Nonzero b_a is needed for conditional subproblems.

Equation (8) is an explicit clarification for this English edition. The archived report uses a recursive XOR description and code snippets that are not identical to this expansion. No claim is made that the archived program already implements the integer-slack construction.

B. Coefficient Convention and Solver Conversion

Let u collect the data and auxiliary bits. We use an upper-triangular convention,

$$E(u) = c + \sum_i Q_{ii} u_i + \sum_{i<j} Q_{ij} u_i u_j. \tag{9}$$

For a single penalty $A(c_a^T u - b_a)^2$, the contributions are

$$\Delta Q_{ii} = A(c_{a,i}^2 - 2 b_a c_{a,i}), \tag{10}$$

$$\Delta Q_{ij} = 2A c_{a,i} c_{a,j} \quad (i < j), \qquad \Delta c = A b_a^2. \tag{11}$$

The channel LLR adds L_i to the diagonal of the corresponding data bit. If a symmetric matrix is used with $u^T Q u$, each off-diagonal polynomial coefficient must be split equally between its two entries. Doubling these entries changes the objective.

Kaiwu SDK 1.1.2 provides QUBO-to-Ising conversion and spin post-processing routines [8]. Its documented matrix conversion introduces an additional reference spin. All variable ordering, reference-spin handling, and spin-to-binary conventions must be preserved when reconstructing a candidate codeword. The decoded word contains the n data variables, not the slack variables or the reference spin. Energy consistency should be checked on small enumerated instances before communication-performance evaluation.

C. Penalty Selection and Adaptive Weights

Penalty weights determine the tradeoff between likelihood and constraint violation for a heuristic solver. A useful sufficient condition can be stated independently of any empirical tuning. If the target syndrome is feasible and

$$\min_a A_a > \sum_i |L_i|, \tag{12}$$

every global minimizer of (8) satisfies all parity constraints. Any infeasible binary assignment has at least one integer residual with magnitude at least one, whereas the total range of the linear channel objective is at most $\sum_i |L_i|$. A feasible assignment thus has lower energy than every infeasible assignment under this sufficient bound.

This bound guarantees the model's global optimum, not a heuristic solver's ability to find it. Excessively large penalties can also compress the effective channel coefficients after hardware rescaling. Practical tuning should jointly consider feasibility, coefficient precision, search budget, and error rate.

The original report proposed the adaptive heuristic

$$A_a = A_0 \left(1 + \alpha \frac{\bar{d}_a - \bar{d}}{\bar{d}} + \frac{\beta}{\gamma} + \eta v_a\right), \tag{13}$$

where $\bar{d}_a$ is the mean neighboring variable degree, $\bar{d}$ is its global counterpart, γ is linear SNR, and v_a is local LLR variance. It reported $(\alpha, \beta, \eta) = (0.12, 0.15, 0.08)$ in one configuration. These are project settings, not established universal optima. Weights must remain positive; preserving the sufficient bound additionally requires a suitable positive floor.

The archived implementation uses a different normalized degree and reliability weighting rule, including $\tanh(|L_i|/2)$, rather than exactly (13). Moreover, nonuniform reweighting of the channel term changes the ML objective unless it represents a justified likelihood calibration. Penalty adaptation and channel reweighting should therefore be distinguished. The present explicit ML model leaves the channel term in (4) unchanged.

TABLE II
EXPLICIT MODEL SIZE FOR SELECTED REGULAR MATRICES

Family	n	m	Slack bits	B	Rank
$(2,3)$	180	120	120	300	119
$(2,3)$	540	360	360	900	359
$(3,6)$	120	60	120	240	60
$(3,6)$	240	120	240	480	120

Dimensions and GF(2) ranks were checked using the archived matrix generator. Slack counts refer to the explicit model in this edition. They exclude an SDK reference spin.

D. Variable Budget and Interaction Cost

The full model requires

$$B = n + \sum_{a=1}^{m} q_a \tag{14}$$

binary variables before any additional Ising reference spin or platform-specific embedding overhead. For zero target syndrome, a degree-three check needs one slack bit and a degree-six check needs two. Thus $(2,3)$ and $(3,6)$ regular matrices both give $B = 5n/3$ and $B = 2n$, respectively, when all row degrees are exact.

Squaring a check residual creates pairwise interactions among its $d_a + q_a$ variables. The per-check coefficient contribution can therefore be dense even when H is sparse. A rough construction cost is $O(\sum_a (d_a + q_a)^2)$ before merging repeated terms. A small number of variables alone does not guarantee low embedding cost or low numerical precision requirements.

IV. Hybrid Decoding Architecture

A. Classical Initialization and Reliability Selection

The proposed architecture first runs a classical decoder to obtain an incumbent word $x^{(0)}$, a syndrome, and bit reliability estimates. Standard sum-product belief propagation (BP) exchanges variable-to-check and check-to-variable messages; its check update has the form

$$r_{a \to i} = 2 \operatorname{atanh}\left(\prod_{j \in \mathcal{N}(a) \setminus \{i\}} \tanh(q_{j \to a}/2) \right). \tag{15}$$

The variable update adds the channel LLR and incoming check messages excluding the recipient. Numerical clipping and a stopping criterion are required in an implementation. A valid-codeword stopping condition is useful, although a zero syndrome alone does not establish that the transmitted word was recovered.

When initialization fails, the project selects unreliable variables using small LLR magnitude and proximity to unsatisfied

checks. A selected set $\mathcal{S}$ can be grown from these seeds until the QUBO budget is reached. The budget must include the auxiliaries of all affected checks; simply imposing $|\mathcal{S}| \le B_{\max}$ is insufficient.

The archived module named BP attenuates LLR magnitudes on unsatisfied checks rather than implementing these standard message updates. The intended BP stage in this architecture is therefore distinguished from that archived routine. This distinction is needed before interpreting any comparison labeled "BP" as a standard BP benchmark.

B. Boundary-Consistent Subproblems

Freezing the complement $\bar{\mathcal{S}}$ of a variable subset changes the right-hand side of its parity equations. For a zero-syndrome full code,

$$H_{\mathcal{S}} x_{\mathcal{S}} = b \pmod 2, \qquad b = H_{\bar{\mathcal{S}}} x_{\bar{\mathcal{S}}}^{(t)} \pmod 2. \quad (16)$$

For a general target syndrome s, the right-hand side becomes $s + H_{\bar{\mathcal{S}}} x_{\bar{\mathcal{S}}}^{(t)}$ over $\mathbb{F}_2$. All checks touched by $\mathcal{S}$ must be retained, including the fixed outside-bit contribution. Replacing (16) with $H_{\mathcal{S}} x_{\mathcal{S}} = 0$ can reject valid global codewords or admit inconsistent updates.

The local QUBO uses the selected LLRs and the syndrome in (16). If its target is infeasible under the fixed outside bits, the decoder must expand or change the selected set, or explicitly record an infeasible local repair. It should not report convergence merely because a subset of checks is satisfied. Full-word verification remains necessary after each update.

C. Optimization and Candidate Selection

The optimizer generates multiple candidates. Auxiliary variables are removed after conversion to the correct binary convention, duplicate data words are merged, and every candidate is tested against the full parity-check matrix. For the feasible candidate set $\mathcal{F}$, the likelihood selector is

$$\hat{x} = \arg\min_{x \in \mathcal{F}} L^T x. \quad (17)$$

This is ML selection *within the returned feasible list*; it is not a guarantee of global ML decoding. If no feasible candidate is available, the decoder may retain the incumbent or minimize a clearly identified soft score $L^T x + \lambda \|Hx \bmod 2\|_1$, but it must report the remaining violations.

The project also proposes classical refinement after QUBO optimization. Such a stage must actually retain the accepted candidate or use an explicit warm-start mechanism. Restarting a deterministic classical decoder from the unchanged original LLRs cannot communicate the QUBO result to that decoder. Any pseudo-LLR injection would be a separately specified heuristic with its own tuning and validation requirements.

D. Kaiwu Integration and Decomposition

Kaiwu SDK 1.1.2 exports `SimulatedAnnealingOptimizer` in its `classical` module, `CIMOptimizer` in its `cim` module, and the QUBO/Ising conversion functions used in the project. The supplied SDK package confirms that these interfaces exist.

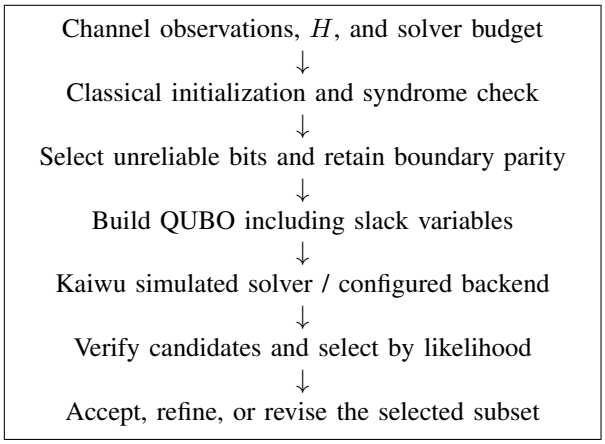


Fig. 1. Proposed hybrid workflow. Boundary consistency, auxiliary-variable accounting, and full-word verification make the intended optimization interface explicit. This diagram describes the architecture, not a claim that every stage has been reproduced end to end.

Its simulated annealer exposes initial temperature, cooling coefficient, cutoff temperature, iterations per temperature, and the size of the returned solution set. This matches the competition's request for an SDK simulated solver. Availability of the API is distinct from successful loading of a particular binary package and from access to a licensed physical backend.

The original report gives illustrative annealing settings of initial temperature 100, cooling coefficient 0.99, and cutoff temperature 0.001, with iteration depth depending on the script. A complete run must record the exact optimizer class, SDK version, parameters, seeds, and whether a fallback backend was used. Calling the classical simulated annealer is not a physical quantum-annealing experiment. The archive also contains a CIM adapter, but retained run identifiers or hardware logs are needed before attributing any particular result to hardware.

For longer codes, the architecture solves a sequence of overlapping subsets under the budget in (14). Each subset retains its boundary syndrome and updates a shared incumbent. Overlap can help repair constraints spanning a partition boundary, while repeated sweeps permit information exchange. Independent parallel optimization is valid only when interactions are coordinated or absent. The archived partition routine processes subsets sequentially; its comments alone do not establish measured parallel acceleration. Decomposition is a resource-management strategy, not a proof of sublinear decoding time.

V. Evaluation Design and Reported Project Results

A. Evaluation Protocol and Evidence Categories

The final Chinese report describes AWGN/BPSK evaluation at 9, 15, and 35 dB, together with broader curves from 0.5 to 6 dB. It discusses nominal lengths 648, 1296, and 2592, adaptive weights, candidate post-processing, and decomposition. These settings and historical values are retained below as the project's

TABLE III
SELECTED HISTORICAL VALUES STATED IN THE FINAL REPORT

Nominal label	SNR (dB)	Hybrid BER	BP BER	Hybrid time (s)
BG1–648	2.0	2.1×10^{-1}	3.2×10^{-1}	0.178
BG1–648	3.0	2.4×10^{-2}	7.8×10^{-2}	0.245
BG2–648	2.0	3.1×10^{-1}	4.2×10^{-1}	0.148
BG2–648	3.0	4.8×10^{-2}	9.8×10^{-2}	0.198
BG1–1296	3.0	1.4×10^{-2}	3.8×10^{-2}	0.486

Entries reproduce the manuscript's stated labels and values; they are not newly measured results. "BG1/BG2" and "BP" are historical labels with the qualifications stated in the text.

TABLE IV
OTHER OPTIMIZATION CLAIMS RECORDED IN THE REPORT

Project item	Historical stated outcome
Degree-aware / channel-aware / combined weights	18% / 25% / 35% improvement
Energy / ML / composite post-processing	8% / 22% / 28% BER improvement
Two / four subgraphs	Nominal maximum length 1296 / 2592
Subgraph timing	0.328 s / 0.612 s
Subgraph BER	0.018 / 0.016

These are original report claims. Retained ablation logs, matched experimental controls, and per-frame evidence were insufficient to verify them. No new ablation result is implied.

TABLE V
HISTORICAL BER CLAIMS AT THE PROBLEM'S SPECIFIED SNRS

Original configuration label	9 dB	15 dB	35 dB
$(2,3)$, 180 input bits	0.024	0.0031	4.2×10^{-5}
$(2,3)$, 240 input bits	0.019	0.0025	3.4×10^{-5}
$(3,6)$, 120 input bits	0.018	0.0024	3.1×10^{-5}
$(3,6)$, 180 input bits	0.015	0.0019	2.8×10^{-5}

Reported manuscript entries, retained for completeness. The archive does not supply sufficient trial records or SNR-definition metadata to reproduce these claims.

numerical record. They are separated from properties checked directly for this English edition.

Three evidence categories are used. *Reported values* are entries in the archived manuscript. *Stored illustrative data* are arrays in the retained JSON files and the scripts that generate the project charts. *Verified properties* are results of local mathematical checks or matrix inspection. The first two categories cannot be promoted to independently reproduced measurements without a traceable experiment log.

A complete benchmark should use valid transmitted codewords and explicitly identify whether BER is measured on information bits or on all codeword bits. The codeword BER and frame error rate (FER) estimators are

$$\widehat{\mathrm{BER}} = \frac{\sum_{f=1}^{F}\sum_{i=1}^{n}\mathbf{1}[\hat{x}_{f,i}\neq x_{f,i}]}{Fn}, \qquad \widehat{\mathrm{FER}} = \frac{\sum_{f=1}^{F}\mathbf{1}[\hat{x}_{f}\neq x_{f}]}{F}. \tag{18}$$

Information-bit BER requires a specified encoder and information-bit extraction map. A random unconstrained binary vector is generally not a valid LDPC codeword. All-zero transmission can be useful for symmetric linear-code tests when the decoding procedure respects the required symmetry; arbitrary channel-dependent reweighting or tie-breaking should be checked before relying on that reduction.

End-to-end timing should include classical initialization, subset selection, QUBO construction, conversion, optimization, candidate processing, and refinement. For remote hardware it should additionally distinguish queueing and communication from physical processing. Model-building time and solve time should be reported separately. Matched channel realizations and independent noise/solver random streams are needed for fair comparisons.

B. Numerical Record from the Final Report

Table III preserves representative entries from the original performance table. For example, the stated BG1–648 row at 3 dB gives a BER ratio of $0.078/0.024 = 3.25$. This is the arithmetic behind the reported improvement, not an independently established gain. The stored data array associates those two BER values with 2.5 dB, whereas its entries at 3 dB are 0.0043 and 0.021. The English edition does not silently substitute one record for the other.

The same provenance distinction applies to the charts. The two retained `complete_experiment` scripts return literal numerical arrays, and those arrays match their respective exported JSON results. The Chinese and English JSON versions also differ in several timing, convergence, and BER entries. No frame-level records were located in the inspected project archive. The figure is therefore retained as an archived visualization of illustrative project data, rather than evidence of a reproduced decoder comparison.

The report further states convergence of 92% for its nominal BG1–648 case at 3 dB, recommends subgraphs of 40–60 variables for BG1-oriented operation and 30–50 for BG2-oriented operation, and lists different base penalties for the two labels. These choices summarize the original design narrative. They are hypotheses or reported configurations pending a controlled evaluation, rather than validated operating recommendations.

C. What the Retained Evidence Can Establish

The rank checks in Table II establish concrete code dimensions. For the archived $(2,3)$ matrix with $n = 180$, rank 119 gives $K = 61$ and $R = 61/180 \approx 0.3389$, rather than 180 information bits or a rate of 0.6. The $(3,6)$ matrix with $n = 120$ has rank 60 and dimension 60. At $n = 240$, its dimension is 120. For the checked $(2,3)$ matrix with $n = 540$, the dimension is 181; a design-rate calculation alone would suggest 180. These are matrix-specific results, not a universal rank statement about the ensemble.

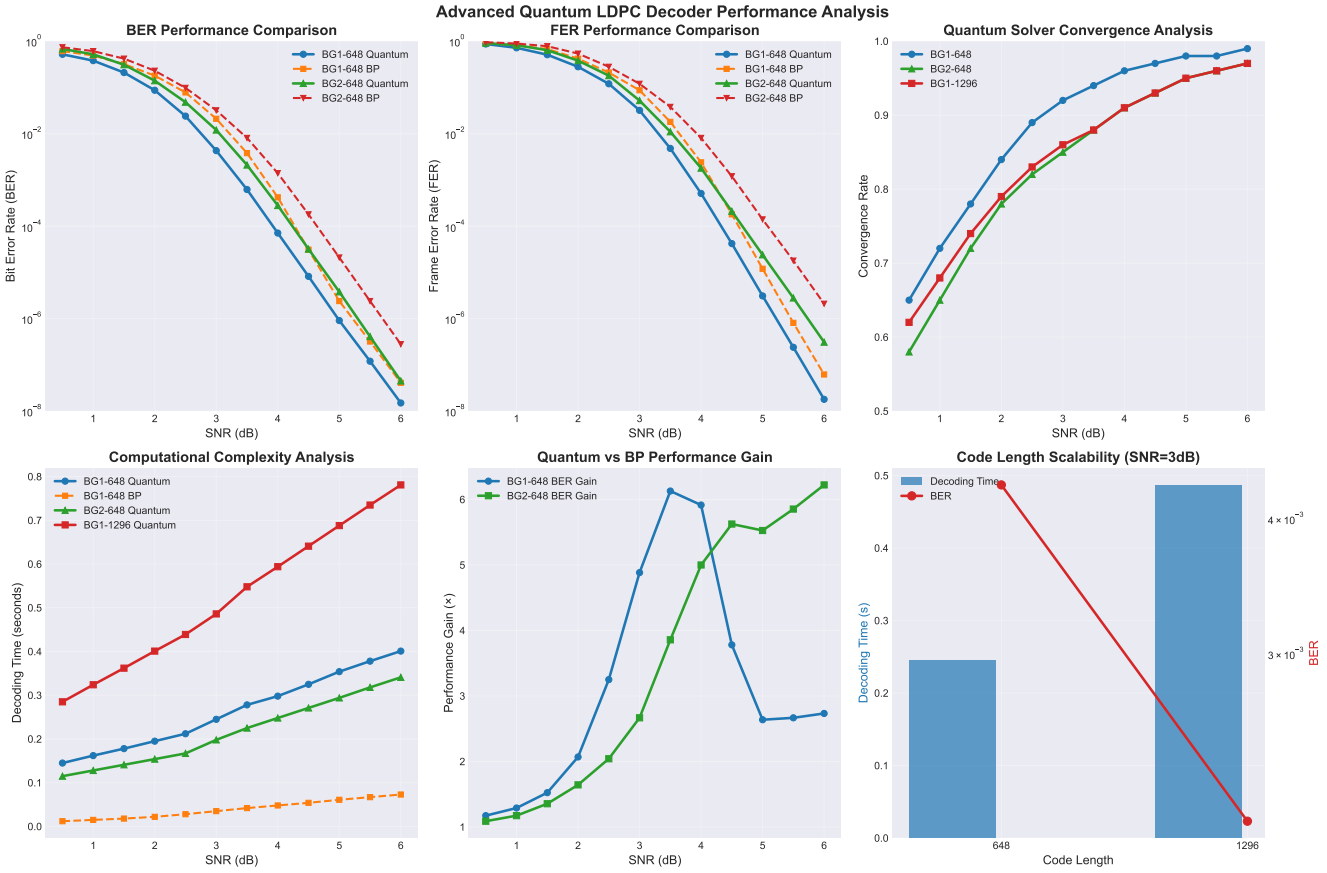


Fig. 2. Archived six-panel visualization from the final project report: BER, FER, convergence, timing, nominal performance ratio, and nominal length scaling. The figure is reproduced unchanged as part of the historical record. Its arrays are explicitly specified in the retained chart-generation script; no frame-level logs were found to establish these curves as measured benchmarks. The embedded terms "quantum," "BP," and BG1/BG2 are original labels and do not independently identify physical hardware, a standard BP implementation, or standardized NR matrices.

The lifted-matrix labels also require care. The archived simplified construction with $Z = 27$ produces 1836 columns for a 46×68 base shape and 1404 for a 42×52 shape. These are not 648-column matrices. Similarly, lifting BG1's full base shape by 54 produces 3672 columns. Code length must be read from H, not from an experiment's label.

Local mathematical checks were added solely to verify the explicit formulation in this edition. Exhaustive enumeration of checks with degrees one through eight and target parities zero and one covered 1020 data assignments; minimizing over slack bits gave the correct parity penalty in every case. For a specified eight-bit, four-check example with 16 valid words, 64 seeded random LLR instances gave identical decisions from exhaustive QUBO minimization and direct codeword ML search. The boundary-syndrome identity was also checked on all 256 words of that example. These tests verify algebra and small-instance consistency, not long-code BER, SDK solver quality, or physical speedup.

D. Statistical and Timing Limits

Very small reported error rates require sufficient trial counts. For example, 200 frames of length 648 yield 129600 decoded bits and a smallest nonzero empirical codeword BER of approximately 7.72×10^{-6}. Values near 10^{-8} cannot be obtained as a nonzero error count divided by that sample size. If no errors are observed, the result should be reported as zero observed errors with the trial count; under an independent Bernoulli bit model, a rough one-sided 95% upper bound is $3/(Fn)$. Within-frame error dependence may require a frame-based uncertainty analysis instead.

The original solver table lists 43.1 ms for classical simulated annealing and 1427.6 ms for the Kaiwu-labeled solve. The latter is about 33.1 times longer in that table. Consequently, those entries cannot support a time speedup. Nor is a percentage energy improvement meaningful without a shared objective, normalization, offset, and verified decoded-bit mapping. Comparisons should report feasible likelihood and error counts in addition to optimization energy.

VI. Discussion and Further Validation

The project's main value is its integrated modeling approach: parity and likelihood are expressed in an optimizer-compatible objective, while classical processing manages reliability estimates and limited problem size. The explicit formulation clarifies when an optimum corresponds to a valid ML solution and how a local update must respect fixed outside bits.

Candidate post-processing provides a practical interface for imperfect optimizers, but it can select only among candidates actually returned.

Several implementation details need reconciliation before an end-to-end performance claim can be reproduced from the retained snapshot. Some demonstration and test scripts import a generator interface absent from the retained LDPC module. The hybrid post-processing code expects result attributes that differ from the project's own result wrapper. Local sparse-matrix indexing and information transfer after a QUBO update also need consistent handling. These observations concern the inspected local snapshot; they are separate from the existence of the Kaiwu APIs, which was confirmed from the supplied version 1.1.2 package. They do not establish that the author's original competition environment had the same state.

An appropriate next evaluation would fix the codeword and information-length interpretation, retain the exact H matrices and encoder map, record the SNR convention, and compare a genuine BP baseline with an explicitly configured SDK simulated solver on identical channel samples. Penalty adaptation, candidate count, and decomposition should be varied one factor at a time under matched time or sample budgets. Each run should retain error counts, seeds, full-word feasibility, timings, backend identity, and any fallback status. Physical CIM evaluation, if performed, should be documented separately from simulation.

These requirements limit the interpretation of the historical results. The archive supports a description of the proposed architecture and its displayed numerical record. It does not establish standards-compliant 5G NR decoding, production deployment readiness, a measured quantum advantage, or an experimentally verified sublinear complexity law. These remain subjects for future verification rather than conclusions of this English edition.

VII. Conclusion

This paper presents an English account of a competition project on QUBO-based hybrid LDPC decoding for coherent optical optimization. It retains the original design themes of channel-aware modeling, unreliable-bit selection, repeated optimization, candidate post-processing, and decomposition. A clarified integer-slack model gives an exact quadratic representation of parity constraints, and boundary syndromes make conditional subproblems consistent with the full codeword. Resource counts and matrix ranks distinguish solver size, codeword length, and information dimension. The historical numerical record is preserved with explicit provenance; small-instance checks establish mathematical consistency of the clarified formulation. Further controlled simulation and documented hardware runs are required to establish comparative decoding performance and any computational advantage.

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